Federated Learning-Assisted Distributed Intrusion Detection Using Mesh Satellite Nets for Autonomous Vehicle Protection

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***Abstract*—The widespread use of intelligent consumer electronics, specifically autonomous vehicles, has exponentially increased. The key enablers of this pervasive are the Internet of Things (IoT), Artificial Intelligence (AI), and Satellite communications, which provide consumers with highly precise and reliable self-driving vehicles. However, autonomous vehicles come also with significant cybersecurity concerns. Attackers can easily use satellite links to launch cyberattacks against autonomous vehicles. An Intrusion Detection System (IDS) is one of the most effective mechanisms for providing secure autonomous vehicles. However, existing IDSs based on machine and deep learning train their models in a centralized server, which uploads data or parameters to the central server for training. This structure of IDS has challenges with vehicle mobility, brings processing delays, and increases privacy and security risks, affecting vehicles' performance. Therefore, for the first time, this paper proposes a new federated learning-assisted distributed IDS using a mesh satellite net to protect autonomous vehicles. We construct a local model using a deep neural network. Then, we provide a mesh federated learning approach that keeps model training local and lets satellites exchange their parameters in a privacy-preserving way. The simulation results show that our proposed model works well while keeping the computation cost reasonable.**

***Index Terms*—Consumer electronics, autonomous vehicles, LEO satellite, federated learning, intrusion detection.**

I. Introduction

WITH the unprecedented deployment of Internet of Things (IoT) technologies and the advent of satellite communications and 5G/6G cellular networks, the digitization of consumer electronics has exponentially increased. One of these intelligent consumer electronics is autonomous vehicles [1], where the idea of self-driving vehicles sharing the road has recently become a reality. High-precision satellites offer the accuracy, ubiquitous connectivity, and reliability that vehicles need to be self-driving [2]. Hundreds of satellites in Low Earth orbit (LEO) and Geostationary Orbit (GEO) work together to form satellite constellations, which alter the static topology of traditional terrestrial networks entirely and allow for flexible, autonomous vehicles deployment [3], [4]. LEO satellites are the most dominant and increasingly used to connect directly with autonomous vehicles as they operate in a lower orbit with less latency and provide wideband Internet access [5], leading to global coverage, low signal delay times, and low signal loss. This provides consumers with safer and more accurate autonomous vehicles than ever in remote or low-connectivity areas [6].

However, one of the key concerns associated with autonomous vehicles is cybersecurity [7]. Attackers could code and make their own keys to get into the car or tamper with the vehicle's critical parts, such as engines. Recent demonstrations have shown that attackers could use malware to control the vehicle via LEO satellites and issue remote commands to unlock it, disable the alarm system, and start the vehicle engine [8]. Therefore, techniques to protect autonomous vehicles and cybersecurity solutions applied in the autonomous vehicles-satellite networks should be updated to reflect the significant advancements in attacker capabilities and the characteristics of the autonomous vehicles and LEO satellites [9], [10]. Intrusion Detection System (IDS) is considered one of the critical parts of the first line of defence in any cybersecurity framework [9], [11].

Classical machine learning algorithms, such as decision tree, random forest, and support vector machine [12], [13] and other common Deep Learning (DL) algorithms, such as a Recurrent Neural Network(RNN), have been widely used to build IDSs for satellite communications with intelligent consumer electronics [14]. Among various DL algorithms, Deep Neural Networks (DNN) seem to have many potentials [15]. DNN-based models for detecting cyber-attacks in autonomous vehicles [16] were also proposed, which can accurately differentiate between benign/normal and malicious traffic. However, IDS-based on DL necessitate powerful computing capabilities in autonomous vehicles. Not to mention that training models with higher complexities on an autonomous vehicle can take time and effort. This centralized training task in the autonomous vehicle or even edge server places a computational load on the central node, which attackers can also compromise [17], [18], making implementing these centralized IDS on the autonomous vehicles or edge server impractical. Therefore, training deep and machine learning model to protect autonomous vehicles while

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not leaking the privacy of training data is still a critical problem.

Several approaches proposed in the literature showed that Federated Learning (FL) could deliver training tasks to intelligent consumer electronics, in particular, autonomous vehicles in the intrusion detection field, without leaking training data privacy [11]. All autonomous vehicle nodes can build a global model without revealing the data. In traditional FL, each node synchronously shares its parameters with a central server, aggregating them and producing new parameters that are sent back to the connected participants/nodes [19]. After the model aggregation, a key issue is with the safe storage of the global model and the parameters at the edge/cloud server. The centralized server often has some risks of tampering, privacy leakage, and single point of failure [17], [20]. In addition, these existing and traditional FL models for IDSs inherit the same drawbacks and limitations of client-server network architecture in terms of scalability, efficient performance, limited vehicle resources, latency and bandwidth, making them inefficient for autonomous vehicles.

Therefore, we suggest using satellite nodes to build IDS for autonomous vehicles. Specifically, we propose a new paradigm that distributes the intrusion detection tasks among a group of LEO satellites in a mesh architecture (Mesh Satellite Nets) instead of sending all parameters or data to a central server or deploying the detection models in the vehicle. This integrates efficient training and asynchronous model sharing among satellite nodes in a mesh network by federated deep learning to ensure reliable and resilient detection systems. By leveraging satellite-based monitoring, cyber-attacks against autonomous vehicles can be quickly and early detected, improving safety and security. The proposed model collaboratively and automatically detects the attacks in a privacy-preserving manner. The main contributions are as follows:

1) We propose a FL-assisted distributed IDS using a satellite mesh net for protecting the autonomous vehicle against cyber attacks. The proposed model provides more robustness and better privacy and security. To the best of our knowledge, this is the first paper to present an FL-based IDS in a mesh architecture using satellites for intelligent electronic consumers (i.e., autonomous vehicles).
2) We present a new learning framework that employs FL and DNN to enable collaborative mesh learning for LEO satellites and enhance attack detection in autonomous vehicles.
3) We perform many experiments using three intrusion datasets to determine the efficiency of our proposed model.
4) The performance of the proposed model has been evaluated and compared with existing FL-based IDSs using many performance metrics.

The remainder of this work is organized as follows. Section II presents the related work, and Section III provides the system model and architecture. Section IV introduces the proposed federated learning-assisted distributed intrusion detection using a mesh satellite net. Experimental evaluation is introduced in Section V. Section VI concludes the work.

II. Related Work

Mesh satellite net-based FL-oriented communication, one of the main enablers for AI-enabled human-centric consumer applications, has the potential to offer advanced vehicle protection services. Intelligent autonomous vehicles that are human-centric and use the FL intrusion detection approach can improve consumers' safety and are crucial for protecting consumers or drivers in dangerous environments. Therefore, researchers have recently focused on developing IDS-based on FL to detect attacks against human-centric consumer applications. For example, Liu et al. [11] presented an FL model for protecting vehicular edge computing against attacks. They used DL to power many detection models distributed at the vehicular edge and the base station layers. In the study of [21], the authors used the client-server FL with software-defined network architecture to secure vehicles in the IoT environment. In related work, Yang et al. [22] used the client-server FL and ConvLSTM algorithm with a node selection mechanism for autonomous vehicles. The experimental results showed the effectiveness of their proposed frameworks in terms of attack detection rate. Abou El Houda et al. [20] proposed a detection model for protecting IoT applications using FL and Game theory. The authors used the non-cooperative game to ensure the provisioning of the required virtual resources to deal with the detected attack. Li et al. [3] developed a distributed network IDS for protecting satellite-terrestrial integrated networks from DDoS attacks using a CNN and FL approach. While Yazdinejad et al. [23] used the FL and cluster-based machine learning algorithms for anomaly detection in blockchain-based IoT systems. However, all the models mentioned above are based on client-server architecture. Privacy and security issues still exist as they use a central server for parameter aggregation, and this server could become a bottleneck for the whole network with increasing the number of connected nodes. In addition, the robustness, latency, bandwidth, and scalability of client-server FL architecture are challenging, especially for autonomous vehicles which connect directly with the LEO satellites.

Many collaborative-based IDS are worth mentioning here to understand the current status of all significant IDSs for mesh architecture. For example, Arshad et al. [24] proposed a collaborative approach by correlating the events generated by multiple intrusion detection models deployed in edge nodes and routers. The proposed approach mainly depended on statistics and threshold for correlating these events collected in a central server, making them highly vulnerable to false alarms and single-point failure issues. Similarly, Wu et al. [25] presented a "Paradise", a new real-time distributed IDS for IoT systems. They exploited the critical features of provenance graphs to analyze the dependency of events generated by multiple IDSs and construct new vectors. In addition, a Cerebellar Model Articulation Controller-based IDS (CMACIDS) was proposed by Kumar et al. [26]. In their proposed IDS, the authors used reinforcement learning and b-spline fit for learning and system adaptability. Their proposed model was evaluated and obtained a good precision of attack detection. Although these models mitigate the drawbacks of client-server architecture, they suffer from

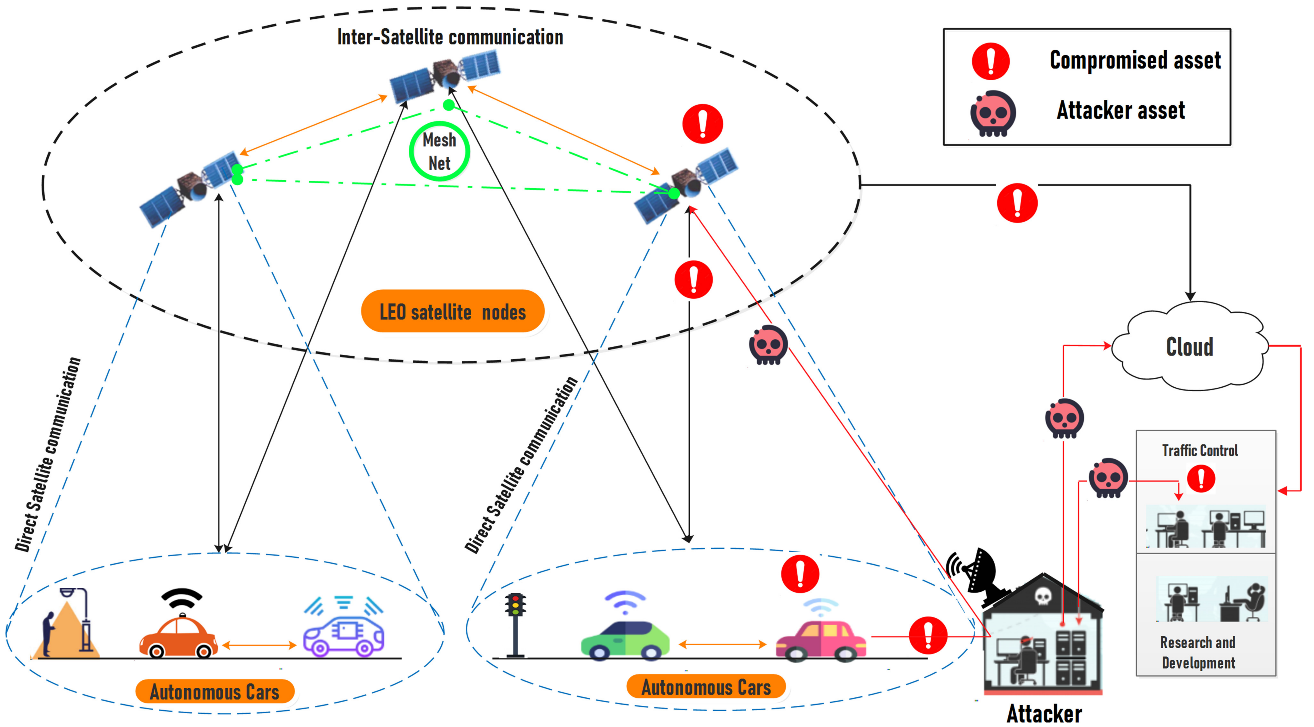


Fig. 1. A mesh satellite net and the threat against autonomous vehicles.

security and privacy challenges that need further studies to secure.

The above discussion shows several intrusion detection models in the literature, each of which has pros and cons. However, this paper proposes a mesh satellite net-based FL-assisted distributed intrusion detection for autonomous vehicle protection. Our proposed model differs from what has been presented in the literature 1) it presents the idea of protecting autonomous vehicles using satellite nodes and distributing intrusion detection tasks among these nodes, and 2) it presents a mesh architecture for exchanging training DL model parameters in a privacy-preserving manner.

III. Proposed Mesh Satellite Net for Autonomous Vehicles

The proposed mesh satellite net for the autonomous vehicles' network and the attacker activities against connected vehicles through satellites are illustrated in Figure 1. The system consists of many connected autonomous vehicles that use LEO satellites in their communications and navigation. The LEO satellites form a mesh network (mesh satellite net) and exchange their training parameters asynchronously in the FL approach. In the attack scenario, attackers could exploit the direct satellite communication links to communicate with LEO satellites, create a backdoor and inject malware into the autonomous vehicle. The injected malware communicates back with the compromised PC, which acts as a command & control server in the traffic control office (to avoid detection). With IDS based on mesh architecture and FL, each LEO satellite node can quickly detect attacks against connected autonomous vehicles as it depends on its local aggregation model and without needing to communicate with a central server.

Also, other satellite nodes can easily and quickly learn about the attacks that their peers and autonomous vehicles face in the mesh architecture. Thus, in our proposed model, each node (i.e., LEO satellite) can get a trained intrusion detection model from other connected nodes in the mesh satellite net. It also acts as an IDS and uses the newly collected data from connected autonomous vehicles to further train and update its local model. Then, the node aggregates the network models of multiple nodes to update their current detection model. Each LEO satellite node communicates asynchronously with other network models. It should be noted that each LEO satellite node in the autonomous vehicle network has only one IDS model and aggregates the received network models with its local model.

To ensure the privacy of the local data of each connected node in the network, each satellite node uses its own intrusion detection data to train its local model and then aggregates the parameters of other trained models in the network without disclosing local data, thereby ensuring the privacy and security of their data. Our proposed model is fully decentralized and distributed as multiple LEO satellites in mesh architecture aggregate and update their network models asynchronously.

IV. Proposed Mesh Satellite Net-Based Federated Learning Intrusion Detection Model

A. *Mesh Federated Learning Model*

Federated learning is used to train the DL-based intrusion detection model (i.e., DNN) as described in Algorithm 1. The LEO satellite nodes (V) are the training nodes and aggregators

Algorithm 1 Mesh Federated Learning Procedure

Data: Initial model parameters $(e.g.,\ i, D_i, W_i, m, A_{i|j}, \mu)$
Result: Global model
for $k \leq K-1$ **do**
  Update parameters for each node i
  for $i \in m$ **do**
    Calculate w_i^{k+1} using *Equation* (5)
    Calculate $\underset{w_i|i\in V}{inf}\ \mathbb{E}_x\left[\sum_{i\in V} q_i^k(W_i) + \sum_{j\in M(i)} \rho/2||A_{i|j}w_i + A_{j|i}w_j^k||_2^2\right]$
    for $i \in m$ **do**
      Select randomly $j \in M(i)$
      Pull $(w_j^{k+1}, \tilde{y}_{j|i}^{k+1})$ *from* j
      Calculate $\tilde{z}_{i|j}^{k+1} = (\theta\tilde{y}_{j|i}^k + (1-\theta)\tilde{z}_{i|j}^k$
    end
  end
end

Algorithm 2 Collaborative DNN Learning Procedure

Data: Initial DNN parameters
Result: Trained DNN
for $epoch \in Epoch\_max$ **do**
  Call Model.train()
  for $i, D_i \in training\_data$ **do**
    Calculate $optimizer.zero\_grad()$
    Calculate $output = Model(input)$
    Calculate $L^i(w)$ using Equation (10)
    Calculate $optimizer.zero\_grad()$
    Calculate $Model.backward()$
    Calculate $Optimizer.step()$
    Calculate $Optimizer.update()$
  end
end

for other network models. These LEO satellite nodes collect data from connected autonomous vehicles. The dataset vector of *(ith)* node $V_i (i \in 1, 2, \ldots, m)$, (m is the number of connected nodes in the mesh satellite network), is defined as (D_i). The weight vector trained on (D_i) is represented by (w_i). When training node's network model, we seek to minimize the sum of local-node costs while keeping the model parameters the same across all connected nodes. This optimization problem can be formulated in Equation (1) and is subject to $A_{i|j}w_i + A_{j|i}w_j = 0, (i \in V, j \in M(i))$, where *M(i)* is the number of LEO satellite nodes connected with node (i) which is similar for the whole other connected LEO satellite nodes in our model (i.e., $M(i) = m$), and $A_{i|j} \in \mathbb{R}$ is the constraint parameter for the edge between two nodes *(i, j)*.

$$\underset{w_i|i\in V}{inf} \sum_{i\in V} F_i(W_i) \tag{1}$$

Because each step of the statistic gradient can be interpreted as minimizing a local quadratic function q_i^k with Lipschitz continuous gradient, and the DNN is continuously differentiable, the new value for weight parameters can be calculated using Equation (2) (where μ is a condition that forms the step size). Also, the previous cost function can be reformulated in Equation (3) as a linearly constrained expectation minimization problem which is also subject to $A_{i|j}w_i + A_{j|i}w_j = 0, (i \in V, j \in M(i))$ [27]. To find a robust solution to this constrained minimization problem, we also use the Augmented Lagrangian method with $\rho > 0$.

$$w_i^{k+1} = \underset{w_i}{argmin}\left(q_i^k(w_i)\right) = w_i^k - 1/\mu\nabla F_i\left(w_i^k\right) \tag{2}$$

$$\underset{w_i|i\in V}{inf}\ \mathbb{E}_x\left[\sum_{i\in V} q_i^k(W_i) + \sum_{j\in M(i)} \rho/2||A_{i|j}w_i + A_{j|i}w_j^k||_2^2\right] \tag{3}$$

However, as we have a lifted dual-variable, i.e., $\lambda_{i|j}$ and $\lambda_{j|i}$ associated with the connected nodes that enable asynchronous and peer-to-peer communication in mesh architecture, the cost function in Equation (3) is also reformulated to support this communication and the learning process among nodes in the mesh satellite net (as described in Equation (4) [27]). Where $(\iota_{(1-P)}(\lambda))$ is an indicator function that forces $(\lambda_{i|j})$ and $(\lambda_{j|i})$ to be identical and the permutation matrix *(P)* enables the lifted dual variables between satellite nodes. Also, $(f^k)^*$ is the convex conjugate [27] of sum over the local-node cost $f^k(w) = \sum_{i\in V} f_i^k(w_i) = q^k(w)$.

$$\underset{\lambda}{inf}\mathbb{E}_x\left[\left(f^k\right)^*\left(A^T\lambda\right) + \iota_{(1-P)}(\lambda)\right] \tag{4}$$

However, to reduce the loss function and update each node in the training process, the new weight parameter of the node after receiving the network models can be calculated using Equation (5) and Equation (6) [17], [27].

$$w_i^{k+1} = \left(\mu w_i^k - \nabla F_i\left(w_i^k\right) + \left(\sum_{j\in N(i)}\left(\tau_{i|j}A_{i|j}^T\tilde{z}_{i|j}^k + \rho w_j^k\right)\right)\right. \oslash \left(\mu + \sum_{j\in M(i)} diag\left(\tau_{i|j}\right) + \rho|M(i)|\right) \tag{5}$$

$$\tilde{y}_{i|j}^{k+1} = \left(\tilde{z}_{i|j}^k - 2A_{i|j}w_i^{k+1}\right) \tag{6}$$

Each pair of LEO satellite nodes exchanges and updates its variables/parameters per around *(R)* updates for each connected node in the mesh net (using a pull protocol). These parameters include weights (w_j^{k+1}) and dual variable $(\tilde{y}_{j|i}^{k+1})$. It also use these values to calculate the new value of $(\tilde{z}_{i|j}^{k+1})$ using Equation (7)

$$\tilde{z}_{i|j}^{k+1} = \left(\theta\tilde{y}_{i|j}^k + (1-\theta)\tilde{z}_{i|j}^k\right) \tag{7}$$

B. Collaborative Mesh Satellite Learning

A Deep Neural Network (DNN) algorithm is selected to accomplish the classification task of intrusion detection while protecting data privacy. DNN is a feedforward neural network that consists of an input layer, many hidden layers and one output layer. The input layer maps the multiple inputs (D_i) to a single output. Assuming (X_i) is the input samples of data owner (D_i), a simple DNN model (i.e., input, one hidden, and output layers) can be expressed as the hypothesis $h(X_i, w)$, and trained locally by the data owner (D^i) as described in Algorithm 2.

$$h(X_i, w) = FC1(FC2(FC3(X_i, w_FC3), w_FC2), w_FC1) \tag{8}$$

where $FC1(*)$, $FC2(*)$, and $FC3(*)$ stand for the three fully connected layers and $(w*)$ is the parameters of the DNN

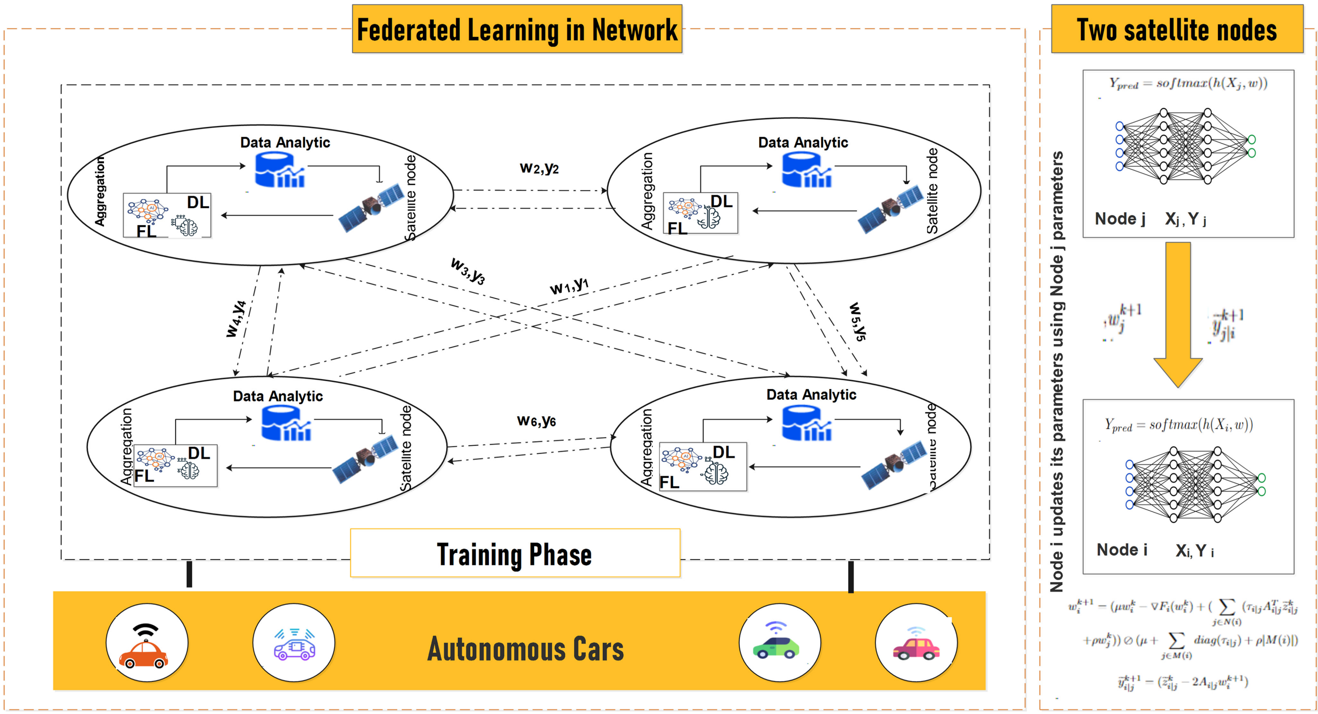


Fig. 2. Training and communication among LEO satellite nodes based on FL in the autonomous vehicle environment.

layers. The output of $h(X_i, w)$ is calculated using a softmax classifier as described in Equation (9).

$$Y_{pred} = softmax(h(X_i, w)) \tag{9}$$

Moreover, we use the negative log-likelihood as a cost function as the proposed model has many attack types. It is computed over a batch of collected observations *(B)* for data owner D_i using Equation (10).

$$L^I(w) = -1/B\left\{\sum_{q=1}^{B}\sum_{p=1}^{cat}(Y^i = P)log\left(\left(e^{w_p^T X^n}\right)/\sum_{q=1}^{B} e^{w_p^T X^i}\right)\right\} \tag{10}$$

Each connected LEO satellite node pulls the network models and then uses the data collected from connected autonomous vehicles to perform local learning to train its local model. The weight w_i^{k+1} and dual variable $\widetilde{y}_{i|j}^{k+1}$ are calculated under the specified data and then this node exchanges and pulls these parameters at random time *(k)* from other network nodes $w_j^{k+1}, \widetilde{y}_{j|i}^{k+1}$. It should be noted that only the encrypted parameters are exchanged among the connected LEO satellite nodes. Figure 2 shows the training and parameter sharing process among the connected LEO satellite nodes in the mesh architecture.

V. Experimental Evaluation

A. Model Evaluation

The third-party library Pytorch was used for the implementation and system model's simulation on the high-performance computer (National Computational Infrastructure (NCI)-Australia) [28]. The datasets used in this experiment include X-IIoTID [29] and Ton-IoT datasets [30] as they represent the modern network traffic and system activities of consumer electronics, including autonomous vehicles in the era of IoT. Also, we use the NSL-KDD [31] as a benchmark dataset for intrusion detection. The X-IIoTID includes more than 421,400 records for normal data and provides 399,417 attack records, which correspond to various new and critical attacks. The TON-IoT dataset consists of 300000 records for normal traffic and 161044 for attacks. The NSL-KDD dataset consists of 77,054 normal and 71460 attack records. The datasets are divided into two forms: 1) The IID data, where data is evenly distributed among the LEO satellite nodes (i.e., they have the same number of classes), 2) The non-IID, where the data is distributed differently, and each LEO satellite node has different data records and classes.

The experiments tested many performance metrics, including loss, accuracy (Acc), precision (P), recall (R), and F1-Score (F1). The experimental parameters were selected based on many experiments where they achieved the best performance (as listed in Table I). We also choose to test 3, 5, and 7 nodes, similar to most studies in the literature. In addition, we evaluate the performance of our proposed model compared with the centralized DNN, where the entire training data is in a central server (e.g., a ground station). We start by assessing the effectiveness of the proposed model's learning. The proposed model's training process is examined by observing how the loss decreases during the aggregation process of the network model. Figures 3 and 4 show the value of loss during the training of both models using IID and non-IID data for NSL-KDD, Ton-IoT, and X-IIoTID datasets, respectively. It is clear that both models tended to converge and reduce the loss with an increasing number of epochs, and both models are

TABLE I
PARAMETERS OF PROPOSED MODEL

Parameter	Value	Parameter	Value
Number of hidden layers and nodes	3 (200 nodes)	Test interval	10
Dropout rate	0.25	ρ	0.5
μ	500	$(\tau_{i\|j})$	3
Batch size	500	Number of epoch	50

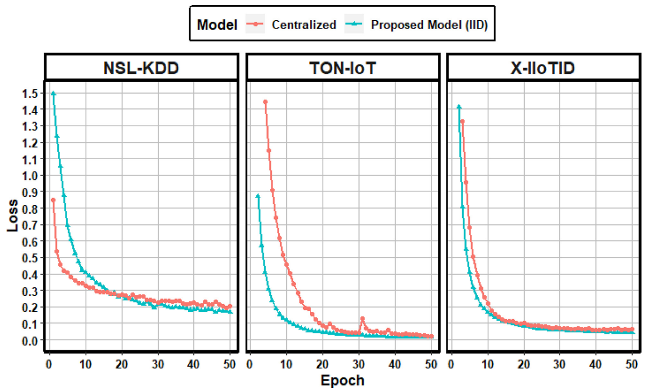


Fig. 3. Loss in the case of IID data distribution.

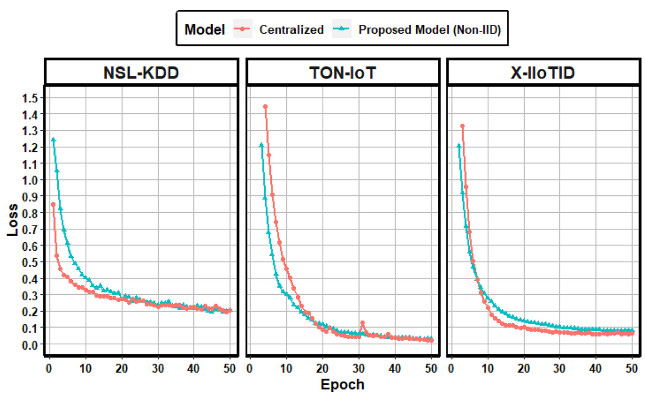


Fig. 4. Loss in the case of non-IID data distribution.

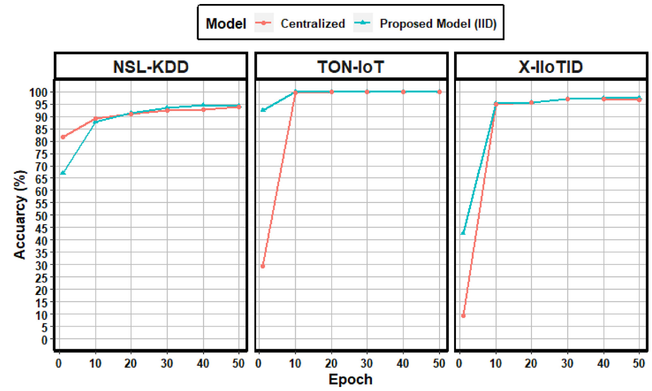


Fig. 5. Accuracy in the case of IID data distribution.

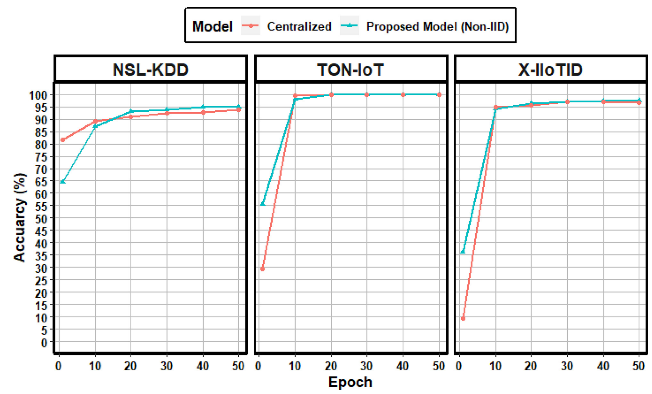


Fig. 6. Accuracy in the case of non-IID data distribution.

equivalent in their loss values. Notably, the proposed model performs well with non-IID data and performs better than the centralized one, particularly for the NSL-KDD and TON-IoT datasets.

The experiment tested the accuracy of models under different epochs. Figures 5 and 6 show that the accuracy increases with the increasing number of epochs. When the epoch is in the range of 15-50, the accuracy slightly increases, and the models start reaching convergence. While for the TON-IoT dataset, the increase is not apparent between 15-50 epochs, it quickly reaches convergence in both figures and for both IID and non-IID data. This is because TON-IoT is an easy dataset and can be understood quickly. Overall, the proposed model performed stably for the non-IID data (as described in Figure 6) and the same results are obtained for IID and for all datasets. The convergence-curve behavior, i.e., accuracy and loss curves, did not significantly change with data distribution among LEO satellite nodes. Therefore, the proposed algorithm is robust against non-IID and IID data, demonstrating its capabilities in dealing with the mobility and heterogeneous data of autonomous vehicles.

Table II shows the detailed performance of the proposed model in terms of Acc, P, R, and F1 under a different number of LEO satellite nodes (i.e., 3, 5, and 7) and for IID data distribution of X-IIoTID, TON-IoT, and NSL-KDD datasets. It can be easily seen that when the number of nodes =3, all LEO satellite nodes (i.e., 1, 2, 3) have approximately the same values for performance metrics and for all datasets. We also obtained the same observation in our experiments with 5 and 7 nodes, where all nodes have identical performance. Also, as the number of LEO satellite nodes increases from 3 to 7, the performance of each model in each node generally improves for the X-IIoTID dataset. For instance, from 3 to 7 nodes, it increased from 96.24% to 97.52%, 97.31% to 98.03%, 96.24% to 97.52%, and 96.74% to 97.70% for Acc, P, R, and F1, respectively. However, the experiments with TON-IoT dataset show that the proposed model achieved the best performance when the number of LEO satellite nodes is 3, and the performance decreases when the number of nodes increases. For example, when the number of LEO satellite nodes is 3, all nodes achieved 100% for all performance metrics. Also, the performance with 5 and 7 LEO satellite nodes ranges between 99.97% to 99.98% for all metrics.

Table III also presents the numerical results of all considered intrusion detection models with non-IID data and varying datasets under a different number of nodes (i.e., 3, 5, and 7). The proposed model generally has a good performance with

TABLE II
THE PERFORMANCE OF OUR PROPOSED MODEL ON IID DATA

Number of nodes	Node Number	X-IIoTID				TON-IoT				NSL-KDD			
		Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
3 Nodes	Node 1	96.38	97.44	96.38	96.78	100	100	100	100	94.48	96.59	94.48	95.25
	Node 2	96.32	97.42	96.32	96.74	100	100	100	100	95.44	96.97	95.44	95.99
	Node 3	96.24	97.31	96.24	96.64	100	100	100	100	94.33	96.67	94.33	95.20
5 Nodes	Node 1	96.67	97.57	96.67	97.00	99.98	99.98	99.98	99.98	93.46	96.28	93.46	94.51
	Node 2	96.85	97.76	96.85	97.19	99.98	99.98	99.98	99.98	93.19	96.39	93.19	94.43
	Node 3	96.87	97.79	96.87	97.21	99.98	99.98	99.98	99.98	92.99	96.24	92.99	94.25
	Node 4	96.96	97.85	96.96	97.30	99.98	99.98	99.98	99.98	93.72	96.47	93.72	94.78
	Node 5	96.66	97.68	96.66	97.04	99.98	99.98	99.98	99.98	93.79	96.37	93.79	94.73
7 Nodes	Node 1	97.49	98.06	97.49	97.69	99.98	99.98	99.98	99.98	94.31	96.59	94.31	95.19
	Node 2	97.36	97.96	97.36	97.57	99.97	99.97	99.97	99.97	94.63	96.77	94.63	95.44
	Node 3	97.42	98.02	97.42	97.63	99.98	99.98	99.98	99.98	94.30	96.81	94.30	95.28
	Node 4	97.51	98.01	97.51	97.68	99.97	99.97	99.97	99.97	94.24	96.57	94.24	95.10
	Node 5	97.31	97.97	97.31	97.54	99.98	99.98	99.98	99.98	94.68	96.72	94.68	95.41
	Node 6	97.52	98.03	97.52	97.70	99.97	99.97	99.97	99.97	94.36	96.58	94.36	95.20
	Node 7	97.29	97.95	97.29	97.52	99.98	99.98	99.98	99.98	94.54	96.83	94.54	95.39

TABLE III
THE PERFORMANCE OF OUR PROPOSED MODEL ON NON-IID DATA

Number of nodes	Node Number	X-IIoTID				TON-IoT				NSL-KDD			
		Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
3 Nodes	Node 1	97.45	97.53	97.45	97.40	99.91	99.91	99.91	99.91	94.93	96.37	94.93	95.44
	Node 2	97.94	98.01	97.94	97.92	99.93	99.93	99.93	99.93	94.27	96.31	94.27	95.03
	Node 3	97.81	97.87	97.81	97.78	99.92	99.92	99.92	99.92	94.27	96.17	94.27	94.95
5 Nodes	Node 1	97.56	97.74	97.56	97.58	99.30	99.38	99.30	99.29	95.42	96.33	95.42	95.81
	Node 2	97.43	97.67	97.43	97.47	99.99	99.99	99.99	99.99	95.66	96.07	95.66	95.82
	Node 3	97.59	97.76	97.59	97.61	99.98	99.98	99.98	99.98	95.63	96.28	95.63	95.89
	Node 4	97.29	97.65	97.29	97.37	99.99	99.99	99.99	99.99	94.71	96.26	94.71	95.33
	Node 5	97.48	97.65	97.48	97.50	99.96	99.96	99.96	99.96	95.60	96.32	95.60	95.88
7 Nodes	Node 1	97.57	97.79	97.57	97.62	99.95	99.95	99.95	99.95	95.03	96.75	95.03	95.73
	Node 2	96.96	97.42	96.96	97.11	99.98	99.98	99.98	99.98	94.68	96.53	94.68	95.41
	Node 3	97.58	97.81	97.58	97.63	99.98	99.98	99.98	99.98	95.34	96.66	95.34	95.86
	Node 4	97.50	97.76	97.50	97.56	99.98	99.98	99.98	99.98	95.24	96.74	95.24	95.84
	Node 5	97.56	97.79	97.56	97.61	99.98	99.98	99.98	99.98	95.24	96.67	95.24	95.80
	Node 6	97.50	97.78	97.50	97.57	99.98	99.98	99.98	99.98	95.53	96.66	95.53	96.02
	Node 7	97.27	97.76	97.27	97.43	98.67	98.96	98.67	98.63	94.89	96.07	94.89	95.28

all datasets and nodes. Importantly, it has stable performance and is similar to the IID data. The experiments with X-IIoTID show that accuracy ranges from 97.27% to 97.96%, and from 97.53% to 98.01%, 97.27% to 97.96%, 97.11% to 97.92% for P, R, and F1, respectively. With the TON-IoT dataset, the Acc and R range from 98.67% to 99.99%, 99.38% to 99.99%, and from 98.63% to 99.99% for P and F1, respectively. Also, the results of the NSL-KDD dataset range from 94.27% to 95.66% for Acc and R while 96.07% to 96.75% and 94.95% to 96.02% for P and F1, respectively.

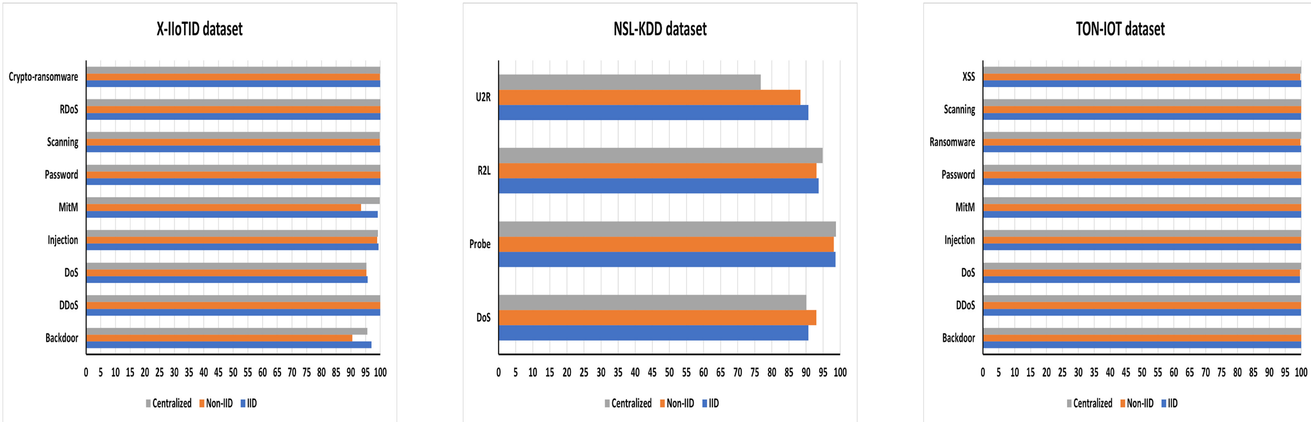
The detection rates for the different attacks and three datasets are shown in Figure 7(a), 7(b), and 7(c). The results include the detection rate for each attack under centralized, IID, and Non-IID data distribution. It can be seen that the performance of our proposed model in classifying different types of attacks in the TON-IoT dataset is approximately identical. This is mainly because of the distinguishable features of each attack in this dataset. For NSL-KDD, the attack detection was a bit challenging due to the minor classes and lack of the appropriate number of attack observations for the training model appropriately. However, most importantly, we can observe that the detection rate of attacks is relatively high for the three datasets, proving our proposed model's efficiency.

TABLE IV
AVERAGE PROCESSING TIME (SECONDS) PER EPOCH AND NODE

Dataset	IID			Non-IID		
	3	5	7	3	5	7
X-IIoTID	23.35	30.05	32.33	23.31	31.68	32.06
TON-IoT	12.84	17	18.04	12.78	18.05	18.85
NSL-KDD	4.99	6.57	7.88	4.92	6.54	7.68

B. Time Complexity Analysis

We calculated the average processing time, including the time of training, transferring, exchanging and updating parameters. Each node exchanges its parameters 220 communication rounds, with 11000 total communication rounds. Table IV shows the calculated value for three datasets. As described in the Table, the average processing time (seconds) per epoch and node is varied between the number of LEO satellite nodes and the data distribution (i.e., IID or non-IID). More specifically, the value of average processing increases with the number



(a) X-IIoTID attack detection rate (b) NSL-KDD attack detection rate (c) TON-IOT attack detection rate

Fig. 7. Detection rate of different attack type.

TABLE V
PERFORMANCE COMPARISON WITH AVAILABLE WORKS

Model	Accuracy	Recall
FedCNN-MLP [19]	88.10%	66.88%
FL-MGVN [32]	82.16 %	57.1%
Our Proposed Model	94.44%	94.44%

of connected nodes (i.e., 3, 5, 7). It ranges from 23.31 to 32.23 seconds for X-IIoTID, 12.84 to 18.85 seconds for TON-IoT, and 4.92 to 7.88 seconds for NSL-KDD datasets. Also, the average processing time for non-IID data is less than for IID for three datasets. The model seems to learn quickly when it has less number of classes. In addition, because of its small size, the NSL-KDD dataset has the least average processing time. Overall, the proposed model consumes a reasonable amount of time, irrespective of the data size and distribution, which makes it suitable for detecting attacks against autonomous vehicles quickly and without any overhead on LEO satellites.

C. Comparison With Available Works

This subsection explores other works comparable to our model and evaluates their accuracy and detection rate/recall performance. We selected L-MGVN [32] and FedCNN-MLP [19] using the NSL-KDD dataset in this comparison. Table V shows that our proposed model achieves high accuracy and recall compared with other models. Both FL-MGVN and FedCNN-MLP have low-performance detecting attacks, ranging from 57.10% to 66.88%. This is because FedCNN-MLP and FL-MGVN models use client-server FL, where the average of parameters of selected connected nodes (not all) is calculated in the server to create a global model, and the selected connected nodes upload their models synchronously. Thus, choosing specific nodes in the training process for the global model could lead to not learning about all attacks. These client-server FL models also have some risks of tampering, privacy leakage, and a single point of failure on the server side. It also has scalability, flexibility, and cost limitations in autonomous vehicles, making them unsuitable for consumer electronics (i.e., vehicles) and even satellite networks. All these weaknesses of the existing models have been resolved in our proposed model using a mesh satellite net architecture.

Our proposed model is fully decentralized and employs a mesh architecture without a centralized server or a need to select participants to provide better flexibility. It can handle heterogeneous data efficiently, as demonstrated in non-IID experiments. It is also able to detect malicious behavior and identify its type efficiently. However, our proposed model has some limitations associated with trust in the connected nodes, necessitating blockchain implementation. Also, selecting the best DNN structure is challenging and may affect the detection model performance. This limitation could be solved using evolutionary algorithms such as a genetic algorithm.

VI. CONCLUSION AND FUTURE WORK

Autonomous vehicles are one of the recent advances in intelligent consumer electronics. While this advent of technology has provided more efficient infrastructure and increased road safety, it has also come with challenges associated with cybersecurity. Therefore, this paper proposed a new detection model that uses a mesh satellite net architecture and asynchronous FL to identify cyber-attacks against autonomous vehicles. Our proposed model mainly focuses on enabling distributed intelligence and decision-making related to attack detection for autonomous vehicles using LEO satellite nodes securely and reliably. Its performance was evaluated using three datasets (including IID and non-IID), demonstrating its efficiency in attack detection.

In future work, we will focus on improving the model performance and studying the effect of implementing the blockchain in the communication and exchange of parameters among LEO satellites on the model performance. We also plan to test the model in a simulated environment of autonomous vehicles communicating with LEO satellites and investigate its performance in such a setup.

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